VISHWAKARMA AWARDS
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VISHWAKARMA AWARDS 2026-27

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Vishwakarma Application ID: VK26-AU-PAI-0352

TEAM NAME: TECHTONICS

TRACK: PHYSICAL AI FOR RESILIENCE AND INCLUSION

MENTOR INSTITUTE: ASHOKA UNIVERSITY

PROJECT TITLE: AGRO-PULSE

TEAM MEMBERS:

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ENCODERLESS SENSOR-FUSION • PRECISION WEEDING & FERTIGATION ROVER

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PROBLEM STATEMENT

NEED FOR AGRO-PULSE

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PROJECT OVERVIEW

Project Name: AGRO-PULSE

Core Innovation: Encoderless, sensor-fusion Physical AI rover for precision weeding and fertigation, with solar-assisted power, closed-loop velocity control, and targeted micro-spraying.

THE EXISTING AGRICULTURAL PROBLEM



AI GENERATED

THE AGRO-PULSE SOLUTION



AI GENERATED

AI VISION
Weed detection

SENSOR FUSION
Encoderless motion estimate

PRECISION SPRAY
Fast valve shutoff

SOLAR + CONTROL
Off-grid, closed-loop drive

TARGET BENEFICIARIES & UNMET NEED

SMALLHOLDER FARMERS
Affordable precision tools

SUSTAINABLE FOOD
Less chemical overuse

RURAL COMMUNITIES
Safer, resilient farming

THE UNMET NEED

An affordable, off-grid rover that targets weeds and crop nutrition, limits unnecessary spraying, and operates reliably in real field conditions.

Excessive chemical usage
Broadcast spraying wastes chemicals when weeds occupy only part of a field.

Soil & water contamination
Unselective spraying can affect soil health, water quality, and food safety.

Farmer health risk
Manual knapsack spraying increases direct chemical exposure.

High cost & low access
Expensive precision machines are often out of reach for small farms.

Wheel slip in real fields
Encoder-only odometry can drift on muddy, uneven furrows.

Need for resilient farming
Smallholders need affordable, solar-assisted precision weeding and fertigation.

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PROPOSED SOLUTION

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AGRO-PULSE

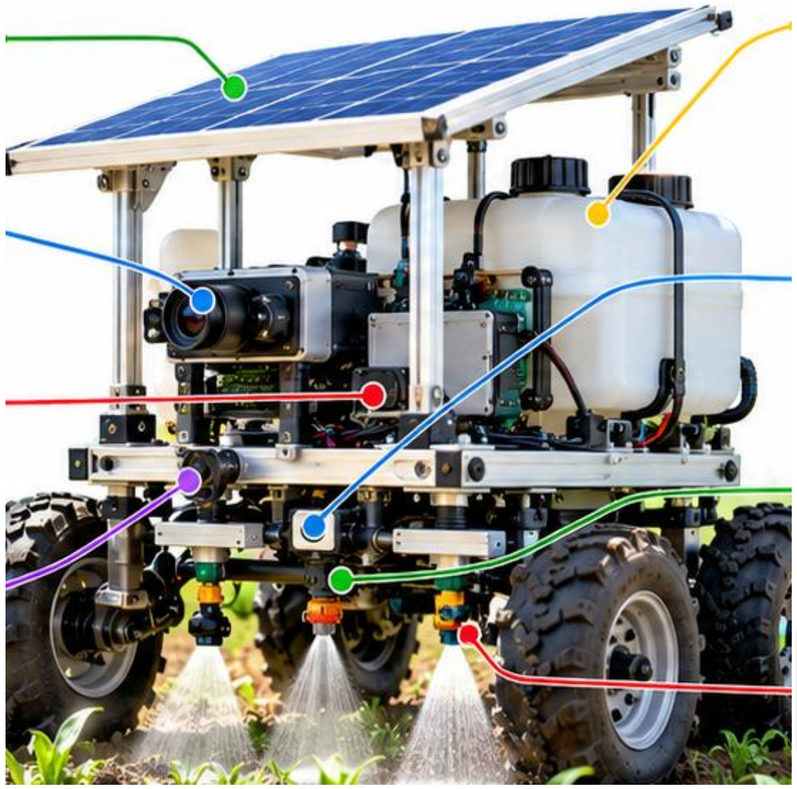
Encoderless Sensor-Fusion Physical AI Precision Weeding & Fertigation Rover with Off-Grid Solar Power and Sub-1.2 ms Valve Shutoff

SOLAR POWER CANOPY
30 W MPPT; off-grid power for 6+ hours

AI VISION CAMERA
OV9281 global shutter for real-time weed detection

STM32G431 CONTROLLER
Sensor fusion, closed-loop velocity and spray control

IMU (BMI160)
Body-motion compensation for stable ground velocity



LIQUID TANK & PUMP
Demand-responsive herbicide / fertilizer delivery

ToF HEIGHT SENSOR
VL53L0X measures ground distance and nozzle height

OPTICAL FLOW SENSOR
PMW3901 estimates ground-relative velocity without wheel encoders

PRECISION SPRAY NOZZLES
Fast solenoid valve; targeted micro-spray

AI GENERATED

"AGRO-PULSE integrates AI perception, encoderless sensor-fusion navigation, and precision micro-spray actuation into an off-grid solar-powered agricultural rover for targeted weeding and fertigation, enabling sustainable and resilient farming for Indian smallholder fields."

INNOVATION & NOVELTY

OFF-GRID SOLAR POWER INTEGRATION
30 W lightweight solar canopy with MPPT charging for extended field operation.

ENCODERLESS SENSOR-FUSION NAVIGATION
Optical flow (PMW3901), IMU (BMI160) and ToF (VL53L0X) support ground-velocity estimation without wheel encoders.

AI-BASED SELECTIVE WEEDING & FERTIGATION
OV9281-based weed detection targets herbicide or bio-liquid fertilizer only at identified locations.

SUB-1.2 ms FAST VALVE SHUTOFF
Fast solenoid actuation is designed to reduce post-spray dripping and chemical wastage.

CLOSED-LOOP VELOCITY & PUMP CONTROL
STM32G431-based control regulates rover speed and demand-responsive pump pressure.

DUAL-MODE OPERATION FOR RESILIENCE
Mode A: precision weeding; Mode B: micro-fertigation for crop nutrition.

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USER JOURNEY

FROM BLANKET SPRAYING TO PRECISION, SENSOR-FUSION WEED CONTROL

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75–85%	~0.30 m/s	₹24,800	30 W
Herbicide waste cited in proposal	Target rover speed	Estimated prototype BOM	Proposed solar canopy

BEFORE | MANUAL BROADCAST SPRAYING



Broad spraying can waste chemicals and expose the operator.

AI Generated

AFTER | AGRO-PULSE TARGETED CONTROL



Camera + sensor fusion + closed-loop control target spray timing.

FARMER USER JOURNEY

1 IDENTIFY	2 ESTIMATE	3 POSITION	4 SPRAY
Camera detects weed	Optical flow + IMU + ToF	Integrate ground distance	Trigger valve at nozzle

TARGET BENEFICIARIES & BEFORE-STATE USER FLOW

TARGET USER

Smallholder farmers (1–5 acres), Farmer Producer Organizations and Custom Hiring Centres.

CURRENT BOTTLENECK

Manual spraying treats broad areas; wheel encoders can misread movement on muddy furrows.

AFFORDABILITY GAP

Commercial selective systems are described in the proposal as costing over ₹1.5 crore.

AGRO-PULSE RESPONSE

Proposed low-cost rover combines weed vision, encoderless motion estimation and targeted micro-spray / fertigation.

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SYSTEM ARCHITECTURE

AGRO-PULSE

Encoderless Sensor-Fusion Physical AI Rover for Targeted Weeding & Fertigation

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1. PERCEPTION & NAVIGATION PATH

Detects weeds and estimates ground motion without wheel encoders.

AI VISION CAMERA (OV9281)
Real-time weed detection
Target localization (X, Y)

IMU (BMI160)
6-DOF motion sensing
Rotation and acceleration

OPTICAL FLOW (PMW3901)
Ground-relative motion (Δx, Δy)
Encoderless velocity estimate

ToF HEIGHT SENSOR (VL53L0X)
Measures ground distance
Height scaling for optical flow

2. CENTRAL CONTROL & SIGNAL PROCESSING

Fuses sensor data, runs AI inference, and controls rover, pump and valves.

RASPBERRY PI 4B
Vision Processing
- AI weed detection
- Target coordinates
- Command data via UART

UART 115200

STM32G431
Control & Actuation
- Sensor fusion
- Closed-loop PI speed
- Pump / valve control
- Safety monitoring

USER INTERFACE (OPTIONAL)
Local Wi-Fi dashboard • configuration • telemetry • data logging

STATUS INDICATORS
Power • operation
Fault indication

POWER / SAFETY
Battery status
Current protection

3. REAL-TIME ACTUATION PATH

Delivers precision micro-spray for selective weeding and fertigation.

A. WEEDING MODE — SELECTIVE HERBICIDE SPRAY

TANK
Herbicide / liquid tank
5–10 L
→
PUMP
Demand-responsive 12 V pump
→
FAST VALVE
Solenoid valve sub-1.2 ms target
→
NOZZLE
Targeted micro-spray
10–15 ms actuation

B. FERTIGATION MODE — MICRO-NUTRIENT INJECTION

BIO-LIQUID TANK
Nutrient solution
5–10 L
→
PUMP CONTROL
Adjustable flow
0–100 PSI
→
SOLENOID VALVE
Precise injection fast shutoff
→
ROOT NOZZLE
Low-volume delivery direct to root zone

4. POWER MANAGEMENT SYSTEM

Provides off-grid solar power and stable rails for field operation.

30 W SOLAR CANOPY
MPPT charging
→
$LiFePO_4$ BATTERY
12 V pack; 6+ h target
→
DC–DC REGULATORS
5 V / 3.3 V rails
→
POWER DISTRIBUTION
Pump, control & sensors

5. ROVER PLATFORM

- Rugged 4-wheel chassis
- Solar canopy
- Camera / sensor mount
- Tank, pump and nozzles

LEGEND

→ Data / communication
→ Control signal / PWM
→ Liquid flow
→ Power distribution

SYSTEM WORKING FLOW

1 SENSE
Camera + sensors detect weeds and estimate motion
→
2 PROCESS
Pi 4B runs AI and sends target coordinates
→
3 NAVIGATE
STM32 fuses sensors and controls rover speed
→
4 ACTUATE
Pump and valve activate at the target
→
5 DELIVER
Targeted spray / fertigation with less waste

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TARGET SPECIFICATION & HARDWARE COMPONENTS

TARGET SPECIFICATIONS

PARAMETER	TARGET SPECIFICATION	SELECTED HARDWARE / PROTOCOL
Vision Inference	12–15 FPS target on 320×320 stream	Raspberry Pi 4B + OV9281 global-shutter camera
Ground Velocity Estimation	Encoderless, ground-relative motion	PMW3901 optical flow + BMI160 IMU + VL53L0X ToF
Velocity Control	~0.30 m/s target speed; closed-loop PI	STM32G431 MCU + motor PWM control
Camera-to-Nozzle Timing	200 mm spatial integration distance	Sensor-fusion velocity integration; adaptive valve trigger
Valve Shutoff	Sub-1.2 ms target shutoff	MOSFET driver PCB + active Zener clamp
Spray Pressure	38–42 PSI target operating range	0–100 PSI pressure transducer + closed-loop pump PWM
Off-Grid Power	30 W solar auxiliary; 6+ h runtime target	Flexible solar panel + MPPT controller + $LiFePO_4$ battery
Prototype Cost	₹24,800 estimated prototype BOM	Itemized hardware and enclosure BOM

SELECTED HARDWARE & PRIMARY USE

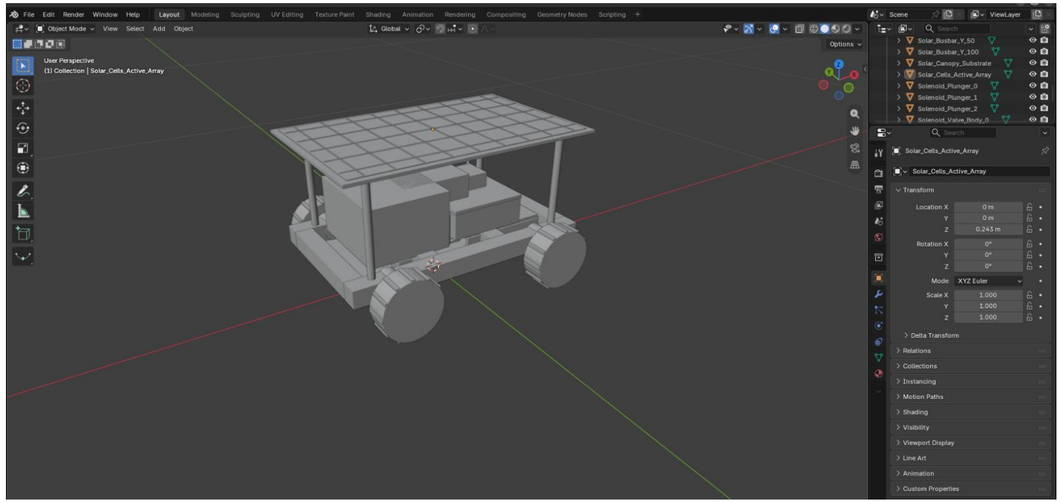
S. NO.	HARDWARE COMPONENT	PRIMARY USE
1	Raspberry Pi 4B + OV9281 Camera	Runs edge weed detection and outputs target coordinates with timestamps.
2	PMW3901 Optical-Flow Sensor	Measures ground-relative motion with reduced dependence on wheel traction.
3	BMI160 6-DOF IMU	Provides chassis motion and rotation data for sensor-fusion correction.
4	VL53L0X ToF Sensor	Measures sensor-to-ground height for optical-flow scaling.
5	STM32G431 Microcontroller	Executes closed-loop PI velocity control and real-time spray timing.
6	12 V Pump + Solenoid Array	Delivers targeted weed micro-dosing and crop-root micro-fertigation.
7	Active Zener MOSFET Driver PCB	Accelerates solenoid current decay for fast valve shutoff.
8	30 W Solar + MPPT + $LiFePO_4$	Provides solar-assisted off-grid power and battery charging.

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CAD DESIGN & MECHANICAL SPECS

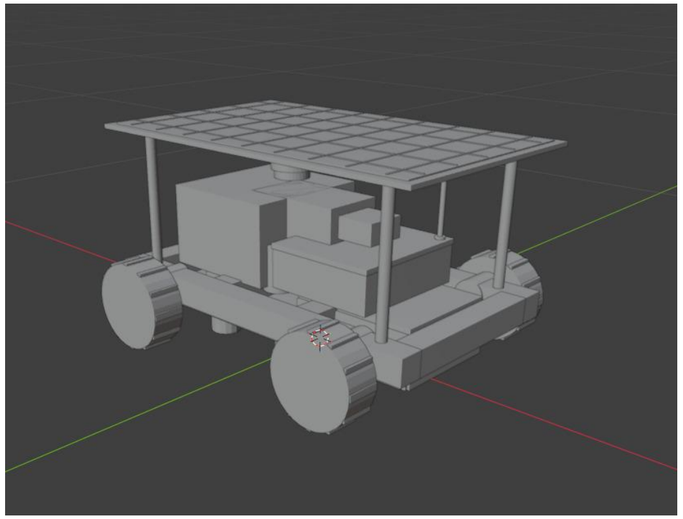
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Assembled CAD Concept: Solar-assisted rover, chassis, tank, camera mast and precision spray system.

ENCLOSURE & INTEGRATION

- **Chassis: Modular aluminium-extrusion frame with four rugged wheels.**
- **Sensor mast: Front-facing camera plus ground-motion and distance sensors.**
- **Fluidics: Removable liquid tank, pump, tubing and individually controlled nozzles.**
- **Power: Battery-powered drive and control; proposed 30 W solar-assist canopy.**
- **Serviceability: Separate electronics enclosure and accessible fluidic module.**



Exploded Assembly: Modular frame, wheel units, solar canopy, tank, pump, sensors and electronics enclosure.

MECHANICAL DESIGN & VALIDATION

Frame & Chassis: Check frame stiffness, payload capacity, ground clearance and fastener retention.

Wheel & Terrain Interface: Evaluate traction, mud clearance, slope stability and chassis vibration.

Fluidic Mounting: Verify tank retention, hose routing, nozzle alignment and leak resistance.

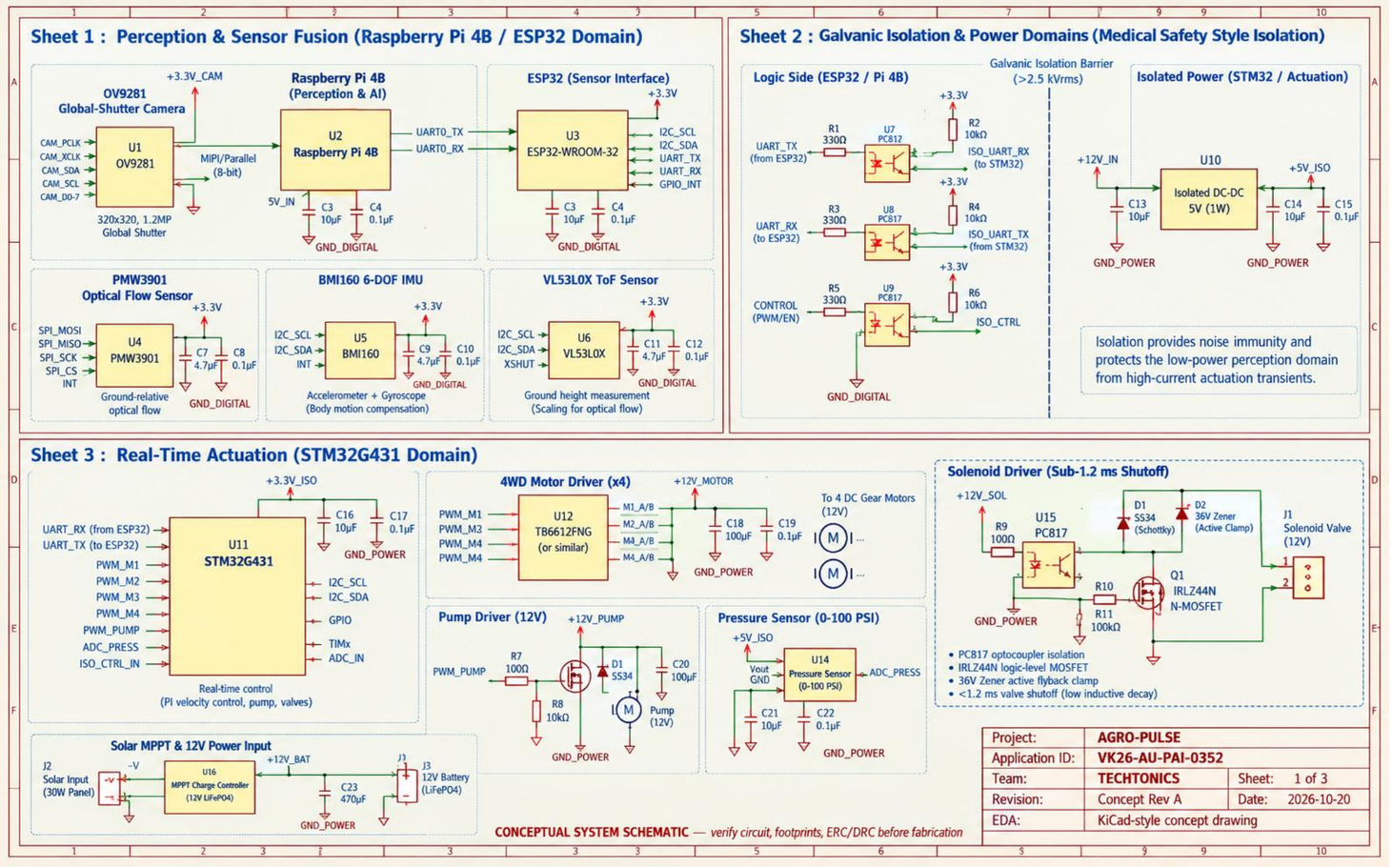
Solar Canopy: Assess panel mounting, centre of gravity, wind loading and rollover risk.

Mechanical Validation: Perform fit checks, vibration tests, controlled drop/impact tests and field trials before claiming durability.

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ELECTRONICS SCHEMATICS & BOM

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COMPONENT	PART / SPECIFICATION	QTY	LEAD TIME (EST.)	COST (INR)
Perception Unit	OV9281 camera + BMI160 IMU + PMW3901 optical flow + VL53L0X ToF	1 set	Local / domestic (3–7 days)	₹4,500
Vision Processor	Raspberry Pi 4B or equivalent edge computer	1	Domestic (5–10 days)	₹6,500
Control & Actuation	STM32G431 + MOSFET / Zener-protected solenoid driver PCB	1 set	Domestic (5–7 days)	₹2,000
Pump & Valve Assembly	12 V demand-responsive pump, fast solenoid valve, precision nozzles	1 set	Local / domestic (3–7 days)	₹3,800
Power Management	30 W solar canopy + MPPT + 12 V $LiFePO_4$ battery + DC–DC regulators	1 set	Domestic (5–10 days)	₹5,500
Mechanical Platform	Four-wheel chassis, mounting frame, tank and enclosure	1 set	Local fabrication (5–10 days)	₹7,500
Total Prototype BOM	Estimated component subtotal; excludes assembly, shipping and contingency	—	—	₹29,800

MAKER JOURNEY



Fig 1: Team members working for stage 2 of Vishwakarma Awards

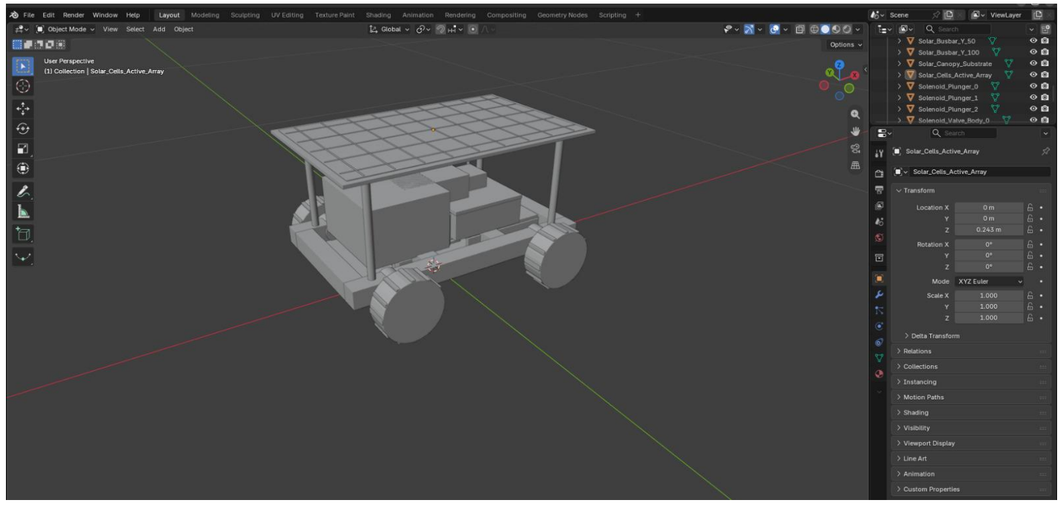


Fig 2: CAD software UI showing active parametric feature tree

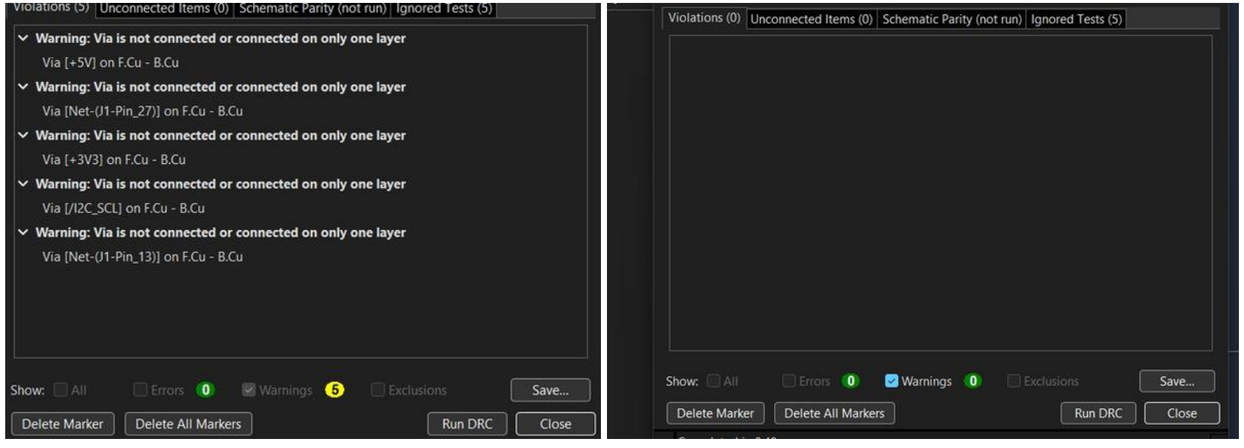


Fig 3: EDA layout software KiCad with DRC rule-check logs.

STAGE 1 TO STAGE 2 ENGINEERING ITERATIONS

- **Design Iteration:** Refining the ROVER design and its structural integrity.
- **CAD Development:** Designing the enclosure and internal component layout.
- **Electrical Design:** Developing sensor interfaces, controller communication, piezo-driver, and valve-control circuits.

GITHUB LINK: https://github.com/TeamTechtonics/Encoderless-Sensor-Fusion-Physical-AI-Precision-Rover

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SAFETY & HAZARD MITIGATION

SAFETY & HAZARD MITIGATION

- **Overpressure Protection:** Regulate the spray line around the proposed 38–42 PSI range; add a pressure relief path and verify pressure limits before field use.
- **Electrical Protection:** Fuse the battery and solar-charging paths; insulate wiring and protect the MOSFET / active-Zener driver against transients and reverse polarity.
- **Chemical & Spray Exposure:** Use enclosed tanks and leak-resistant tubing; keep spray away from the operator and prevent spraying when people or animals enter the treatment zone.
- **Mechanical Safety:** Guard wheels, drivetrain and moving linkages; provide an accessible emergency stop and stable chassis operation on uneven ground.
- **Fault Detection & Fail-Safe:** On sensor failure, UART loss, low battery, abnormal pressure or controller reset, stop the pump, close valves and enter a defined safe state.

STANDARDS & FIELD COMPLIANCE

- **Electrical & Battery Safety:** Assess battery, BMS, fuse, wiring, solar MPPT and charger protection; verify temperature rise and short-circuit response.
- **Electromagnetic Compatibility:** Check whether applicable EMC requirements are met for the MCU, camera, radio/Wi-Fi and motor / solenoid switching.
- **Machinery Risk Assessment:** Use a documented risk-assessment process for moving parts, unintended motion, stability, emergency stop and maintenance access.
- **Ingress & Environmental Protection:** Select enclosure and connector protection appropriate for dust, water splash, mud, vibration and outdoor temperature.
- **Chemical Handling & Usability:** Label the tank and fluid lines, provide safe refill / cleaning procedures, PPE guidance and clear offline status / fault indicators.

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ROADMAP & FAILURE MODE AND EFFECT ANALYSIS

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FAILURE MODE	POTENTIAL HAZARD / EFFECT	PROPOSED FAIL-SAFE / MITIGATION
Wheel Slip / Odometry Drift	Incorrect rover position; target weed may be missed.	Fuse optical flow + IMU; compare motion estimates and stop precision spraying if estimates are invalid.
Weed Detection Failure	Crop may be sprayed or weeds may be missed.	Confidence threshold, region-of-interest checks, and manual stop / override.
Solenoid Valve Stuck Open	Unintended spraying, chemical waste, or crop damage.	Normally-closed valve, pump cutoff, watchdog timeout, and leak / flow checks.
Pump / Line Blockage	No spray or uneven application; pump may overheat.	Flow / pressure monitoring, current limit, and shutdown on abnormal load.
UART / Sensor Data Loss	Stale velocity or target coordinates lead to mistimed spraying.	Message timeout, sensor plausibility checks, and disable spray when data is stale.
Battery / Power Failure	Rover stops in field or controller resets unexpectedly.	Low-voltage detection, battery protection, safe valve-off state, and controlled shutdown.

FABRICATION PHASE (MONTHS 1–2)

October 2026 — Design Finalization

- Refine CAD, nozzle spacing, tank and electronics placement.
- Finalize circuit diagrams, hardware choices and BOM.

November 2026 — Hardware Fabrication

- Assemble chassis, motor drivers, sensor and spray circuits.
- Integrate tank, pump, solenoid and solar-assisted power.
- Verify subsystems and troubleshoot early faults.

INTEGRATION & FIELD DEMO (MONTHS 3–4)

December 2026 — System Integration

- Integrate Raspberry Pi and STM32 over the control link.
- Implement weed detection, motion estimation and synchronized spray actuation.

January 2027 — Testing & Demonstration

- Measure target accuracy, spray timing, power and coverage.
- Test wheel slip, sensor loss, blocked lines and low battery.
- Refine rover and document field-test results.

TECHNICAL CHALLENGES & RISKS

ENGINEERING BOTTLENECKS

- Breath Detection Accuracy: Distinguishing inhalation, exhalation, and pauses using differential pressure signals.
- Real-Time Synchronization: Minimizing communication and actuation delays between ESP32 and STM32G431.
- Piezo Driver Thermal Management: Controlling heat generation during 110 kHz mesh operation.
- PEP Pressure Regulation: Maintaining the proposed 5–15 cmH_2O expiratory pressure range.
- Power Optimization: Operating sensing, control, nebulization, and valve subsystems within the proposed <2.5 W power budget.

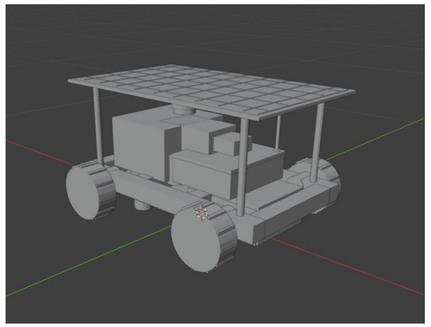
TECHNICAL RISK & PROPOSED MITIGATION

TECHNICAL RISK	PROPOSED MITIGATION
Wheel slip / odometry drift	Fuse optical flow, IMU and ToF; validate on muddy and uneven furrows.
Incorrect weed localization	Calibrate camera-to-nozzle offset and verify target coordinates in field tests.
UART communication failure	Use watchdog timer, timeout handling and a safe actuator state.
Pump / valve overpressure	Add pressure sensing or relief protection and verify during bench testing.
Driver overheating	Thermal monitoring, current limiting and suitable MOSFET / Zener protection.
Battery voltage drop	Regulated power rails, low-voltage detection and controlled shutdown.

ALTERNATIVE SOLUTIONS & FAIL-SAFE STRATEGIES

DESIGN CHALLENGE	PROPOSED APPROACH
Sensor noise / false weed detection	Use image confidence thresholds, filtering and sensor calibration.
Pump or valve failure	Stop spraying, notify operator and enter a safe state.
Incorrect spray timing	Use a deterministic state machine with target-confirmation checks.
EMI affecting sensor readings	Separate sensor wiring from motor / valve power lines and add filtering.
Difficult field maintenance	Use modular sensor, pump, nozzle and battery assemblies.

REFERENCES & SOURCES



URL: Encoderless-Sensor-Fusion-Physical-AI-Precision-Rover/cad at main · TeamTechtonics/Encoderless-Sensor-Fusion-Physical-AI-Precision-Rover · GitHub
Source: Original Parametric 3D CAD Assembly by Team TECHTONICS (Modeled in Blender 5.2.1 | Formats: .blend, .stl, .obj)

GITHUB LINK: **https://github.com/TeamTechtonics/Encoderless-Sensor-Fusion-Physical-AI-Precision-Rover**

IMAGE SOURCES



URL: https://www.naio-technologies.com/en/news/large-scale-vegetable-weeding-in-canada-with-dino/

Source: https://www.naio-technologies.com/